

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Simulate smoothing and sharpening operations on images using frequency domain filters.	
Open Ended Problem 9	Date: 08/09/2026	Enrolment No:92400133050

GitHub Link :

Objective

to implement and evaluate spatial domain image enhancement techniques by applying various smoothing and sharpening filters to corrupted images to enhance their clarity.

Smoothing (Low-Pass Filters)

Smoothing techniques are used to reduce noise and blur the image by suppressing high-frequency details, such as sharp transitions.

Sharpening (High-Pass Filters)

Sharpening techniques enhance image clarity by emphasizing rapid transitions in intensity, restoring detail to blurry or out-of-focus images.

Gaussian Filter (Low-Pass Smoothing)

A Gaussian filter is a linear spatial smoothing technique that applies a weighted average to a pixel's neighborhood to reduce image noise and blur the image.

Function

It acts as a low-pass filter by attenuating high-frequency spatial components (like sharp transitions or noise anomalies).

Advantage

Because the pixel weights taper off smoothly toward the edges of the kernel, Gaussian smoothing preserves image boundaries better than a

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standard uniform averaging filter. It produces a natural-looking blur without introducing severe "ringing" artifacts.

High-Pass Sharpening Filter

A high-pass filter enhances image clarity by emphasizing rapid transitions in pixel intensity—such as edges, lines, and fine textures—while suppressing low-frequency areas of constant brightness.

Function

The filter isolates high-frequency data by calculating the difference between a target pixel and its immediate surroundings. When this high-frequency edge data is added back to the original image, it increases local contrast along boundaries, making the image appear crisper.

Limitation

Because high-pass filters indiscriminately amplify all high-frequency data, they will significantly enhance existing background noise. They are often applied only after an initial smoothing pass has cleaned the raw image.

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Code

```

import cv2

import numpy as np

import matplotlib.pyplot as plt


def gaussian_filter(image, sigma):

    image = np.float32(image)

    dft = cv2.dft(image, flags=cv2.DFT_COMPLEX_OUTPUT)

    dft_shift = np.fft.fftshift(dft)

    h, w = image.shape

    cy = h // 2
    cx = w // 2

    y, x = np.ogrid[:h, :w]

    distance = (x - cx) ** 2 + (y - cy) ** 2

    gaussian = np.exp(-distance / (2 * sigma * sigma))

    mask = np.zeros((h, w, 2), np.float32)

    mask[:, :, 0] = gaussian

    mask[:, :, 1] = gaussian

    result = dft_shift * mask

    result = np.fft.ifftshift(result)

    output = cv2.idft(result, flags=cv2.DFT_SCALE | cv2.DFT_REAL_OUTPUT)

    output = cv2.normalize(output, None, 0, 255, cv2.NORM_MINMAX)

    return np.uint8(output)


def high_pass_filter(image, strength):

    image = np.float32(image)

    dft = cv2.dft(image, flags=cv2.DFT_COMPLEX_OUTPUT)

```

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```

dft_shift = np.fft.fftshift(dft)

h, w = image.shape

cy = h // 2

cx = w // 2

mask = np.ones((h, w, 2), np.float32)

mask[cy - strength:cy + strength, cx - strength:cx + strength] = 0

result = dft_shift * mask

result = np.fft.ifftshift(result)

output = cv2.idft(result, flags=cv2.DFT_SCALE | cv2.DFT_REAL_OUTPUT)

output = cv2.normalize(output, None, 0, 255, cv2.NORM_MINMAX)

return np.uint8(output)

```

```

image_paths = [

    "/content/1.png",

    "/content/2.png",

    "/content/3.png",

    "/content/4.png",

    "/content/5.png"

]

sigma = 40

strength = 40

for i in range(5):

    image = cv2.imread(image_paths[i], cv2.IMREAD_GRAYSCALE)

    if image is None:

        continue

    smooth = gaussian_filter(image, sigma)

    sharp = high_pass_filter(image, strength)

```

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```
plt.figure(figsize=(15, 5))
plt.subplot(1, 3, 1)
plt.imshow(image, cmap="gray")
plt.title("Original")
plt.axis("off")
plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap="gray")
plt.title("Gaussian Smoothing")
plt.axis("off")
plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap="gray")
plt.title("High-Pass Sharpening")
plt.axis("off")
plt.suptitle("Experiment 9 - Image " + str(i + 1))
plt.tight_layout()
plt.show()
```

Output



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Experiment 9 - Image 2



Experiment 9 - Image 3



Experiment 9 - Image 4



Experiment 9 - Image 5



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Observation

Gaussian low-pass filtering with $\sigma = 40$ reduces high-frequency noise and smooths fine textures.

The overall structure of the images is preserved, but very small details and thin edges become slightly blurred.

High-pass filtering with strength = 40 enhances edges, boundaries, textures, and fine details.

Sharpening improves the apparent clarity of the images, but it may also amplify noise and produce slight halos around strong edges.

Smoothing consistently reduces unwanted noise and variations, while sharpening improves edge visibility and detail.

The experiment demonstrates the trade-off between noise reduction and detail enhancement using frequency-domain filters.

The final image quality depends strongly on the selected filter parameters; excessive smoothing causes blurring, while excessive sharpening can make the image appear unnatural.

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Conclusion

we used frequency-domain filters for smoothing and sharpening images. Gaussian smoothing helped to reduce noise and made the images smoother. High-pass filtering made the edges and fine details clearer. We learned that proper filter settings are important because too much smoothing can blur details, while too much sharpening can increase noise. Overall, frequency-domain filtering is useful for improving image quality.